

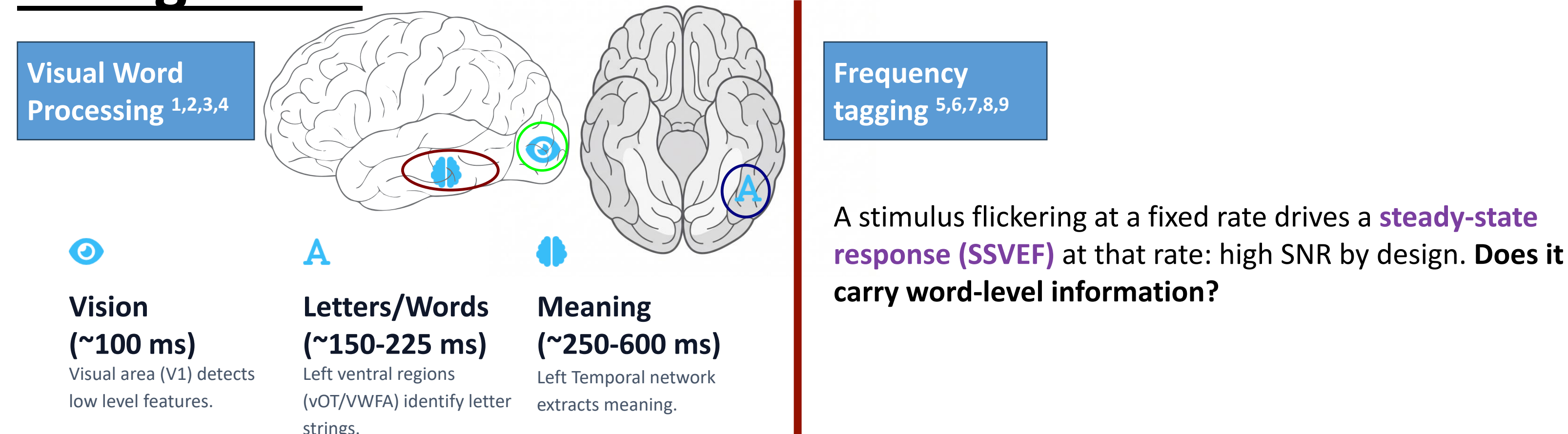
Characterizing word processing pathway using frequency tagging

Kishen Senziani¹, Berk Gercek¹, Nina Kazanina¹; ¹University of Geneva, Faculty of Medicine, Department of Basic Neurosciences

A **6–7 Hz flicker tag** tracks the visual input, not wordness; the **word-locked response** (≥ 250 ms) separates words from consonant strings, maximally in anterior temporal cortex.

Update vs. submitted abstract: with fixation baselines and equal trial counts, the tag effect is null and the source effect is Word > Non-word.

Background



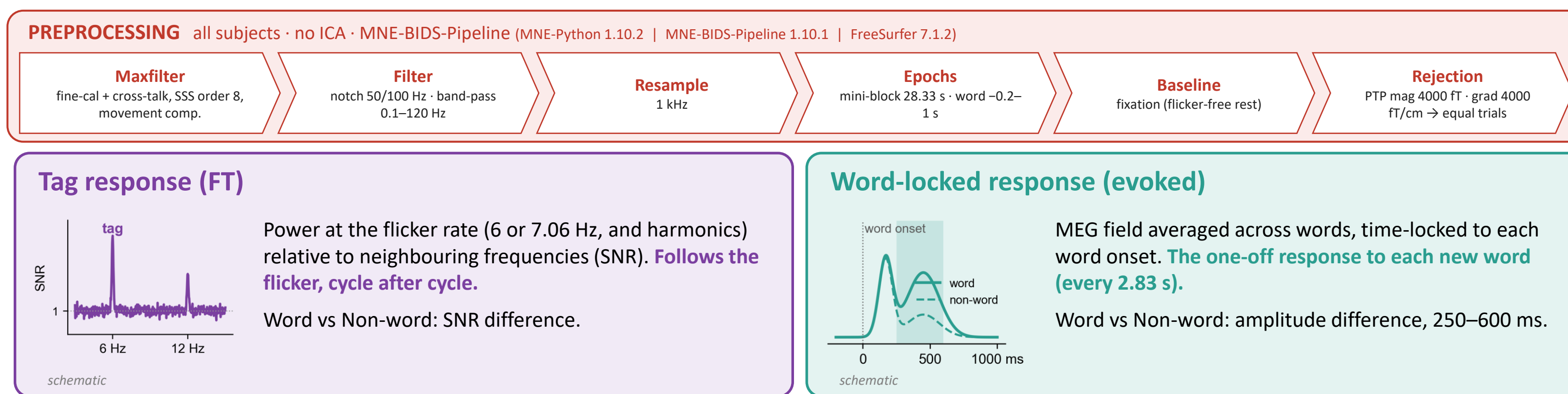
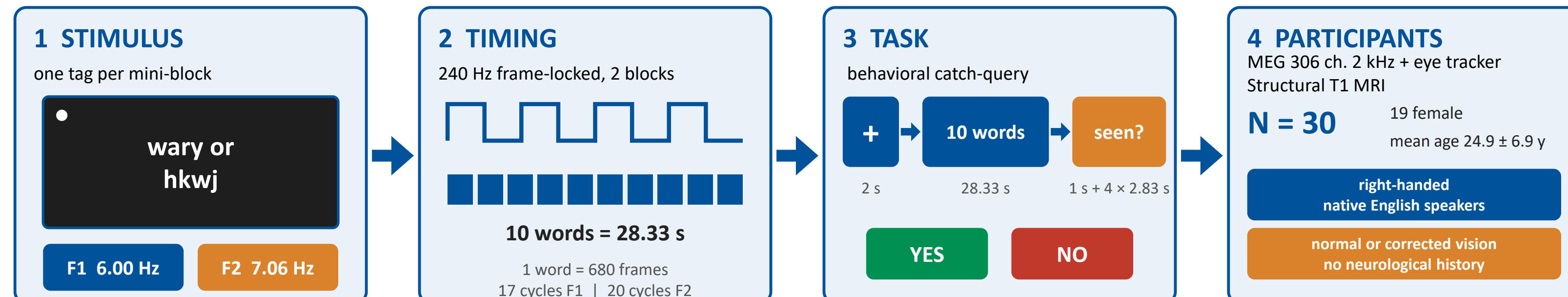
AIM 1

Does MEG source analysis recover the V1 → vOT → temporal sequence of the literature?

AIM 2

Does the tag response (SSVEF) differ between words and consonant strings?

Methods



Pitfalls we hit — and how we fixed them

- Tags off by one FFT bin.** The display ran 0.6 % slow (photodiode: 5.965 / 7.024 Hz): read the SNR at the measured bin.
- Unequal trial counts faked an effect.** A source Word > Non-word effect vanished once counts were equalized.
- No gap between words.** The pre-stimulus window holds the previous word: baseline on fixation instead.
- Check the noise-covariance rank.** After SSS, 3 subjects kept a ~1e-38 eigenvalue (dSPM $\times 10^6$, distorted maps): fixed with `mne.compute_rank(tol='auto')`.
- Match the ink.** Consonant strings had ~10 % more ink and 20 % more height: the occipital tag followed it.



Code · supplementary figures S1–S4 · cohort
github.com/Orion212-lab/snl2026-ft-word-pathway

Cohort 30 recorded; each panel states its N. QC exclusions: no flicker on the photodiode, malformed blocks, no T1 MRI, all epochs rejected; sub-20/24 (source artifact); trigger anomalies sub-10–14 (main results unchanged without them). List via QR.

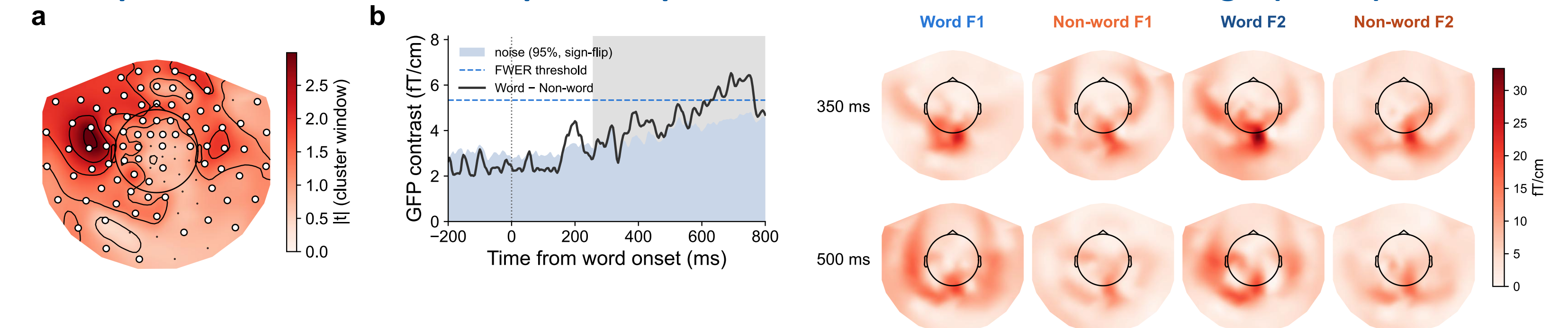
Confirmatory Word vs Non-word at the tag and in the evoked cascade windows (80–130, 150–225, 250–600 ms), left-temporal sensors (a priori). Everything else is exploratory.

Limits consonant strings also differ in orthography, pronounceability and ink · condition-pure blocks · late vOT effect fragile · catch-query accuracy not yet scored

Next pseudowords · item-rate (FPVS) tagging instead of contrast flicker · interleaved conditions · two-word phrase task (intermodulation), already recorded

Results

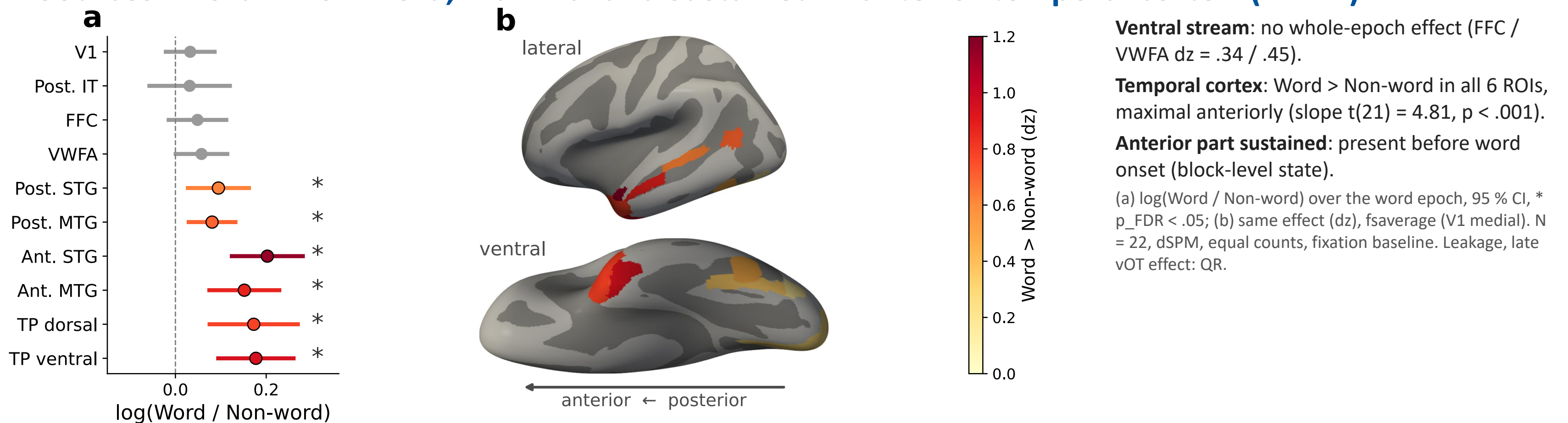
1 · Sensor space: the word-locked response separates words from consonant strings (Aim 1)



One Word > Non-word cluster over left posterior and temporal sensors (N = 26, 84 sensors, $p < .001$), robust to blinks, saccades and artifacts (clean epochs: $dz = 1.31$, 20/21 subjects). Tag $\times$ Wordness: n.s.

(a) mean $|t|$ in the cluster window (N = 26); (b) GFP of W - NW vs sign-flip noise (shaded: 95%; dashed: FWER), exceeded at 628–764 ms; right: grand averages (N = 25) at 350 / 500 ms, ± 25 ms; fixation baseline.

2 · Sources: Word > Non-word, maximal and sustained in anterior temporal cortex (Aim 1)



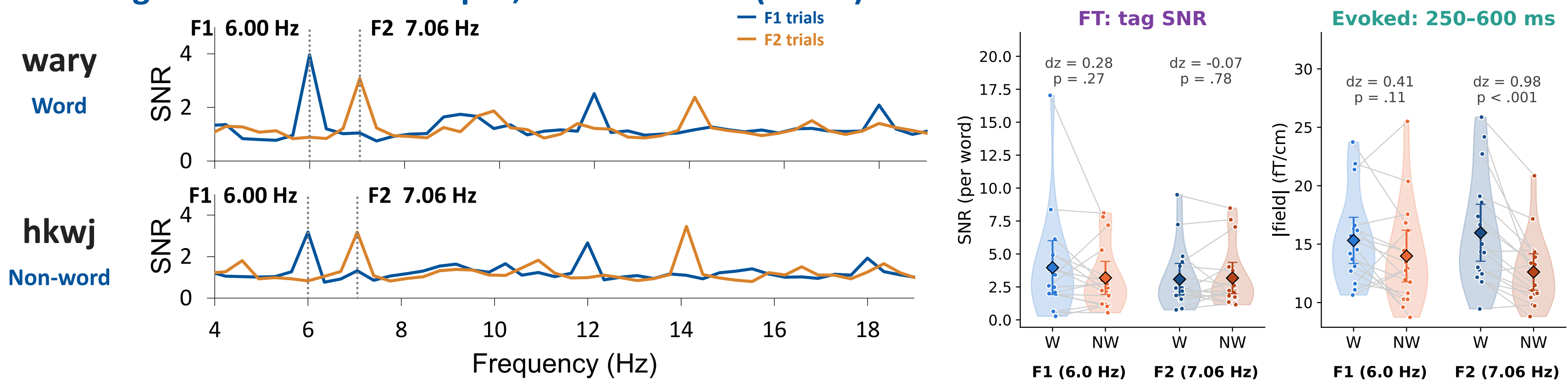
Ventral stream: no whole-epoch effect (FFC / VWFA $dz = .34 / .45$).

Temporal cortex: Word > Non-word in all 6 ROIs, maximal anteriorly (slope $t(21) = 4.81$, $p < .001$).

Anterior part sustained: present before word onset (block-level state).

(a) log(Word / Non-word) over the word epoch, 95% CI, * $p_{FDR} < .05$; (b) same effect (dz), fsaverage (V1 medial). N = 22, dSPM, equal counts, fixation baseline. Leakage, late vOT effect: QR.

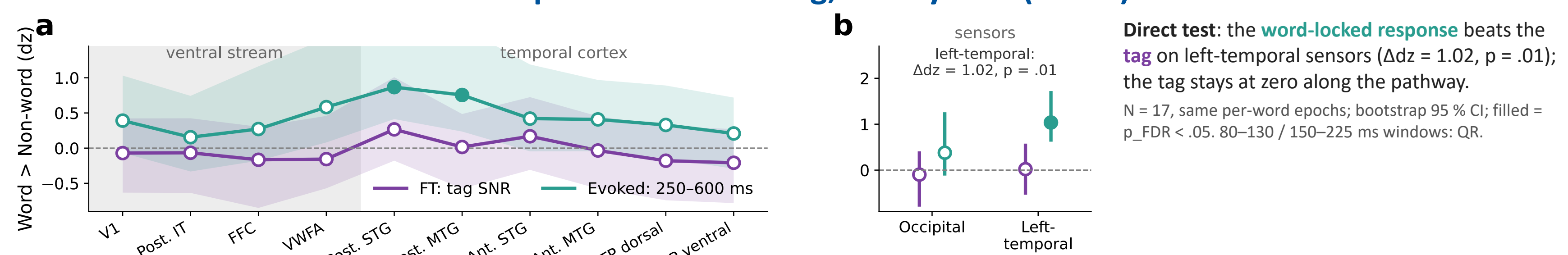
3 · The tag follows the visual input, not wordness (Aim 2)



Tag: Words $\approx$ Non-words over left-temporal sensors (equivalence $|dz| < .5$, $BF_{01} = 4.2$); over occipital sensors Non-words > Words ($dz = -.88$), following their extra ink. **Evoked** 250–600 ms: Words > Non-words ($dz = .98$, $p < .001$).

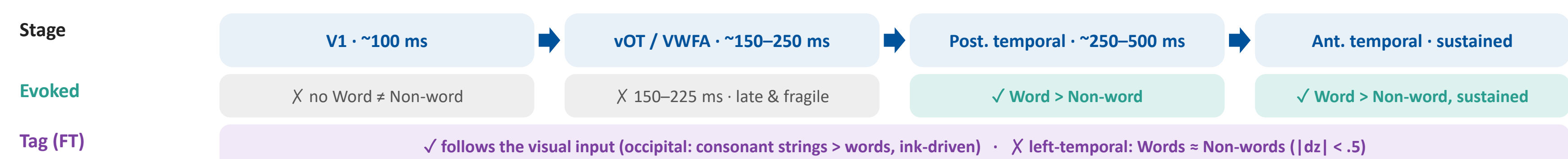
Left: per-word phase-locked SNR (2.83 s FFT, $df = 0.35$ Hz), left-temporal gradiometers. Right: same 17 subjects and epochs for both measures; diamonds = mean $\pm 95\%$ CI. Harmonics / occipital: N = 20.

4 · Direct test: the word-locked response beats the tag, ROI by ROI (Aim 2)



Direct test: the **word-locked response** beats the **tag** on left-temporal sensors ($\Delta dz = 1.02$, $p = .01$); the tag stays at zero along the pathway.

N = 17, same per-word epochs; bootstrap 95% CI; filled = $p_{FDR} < .05$. 80–130 / 150–225 ms windows: QR.



Aim 1 — partly: temporal stages replicated, vOT only late and fragile, latency order n.s. **Aim 2 — no:** the tag indexes visual input, not wordness; the word-locked response does.